

Below is a **MASTER TABLE** of **algorithm types** used in **Java, Python, and AI**, with **software implementation use-cases + real-life examples**.

✓ **MASTER TABLE — Types of Algorithms (Java + Python + AI) with Use Cases & Real-Life Examples**

#	Algorithm Type / Category	What it Solves	Common Algorithms / Examples	Java/Python/AI Software Use Case	Real-life Example
1	Sorting Algorithms	Arrange data in order	Quick Sort, Merge Sort, Heap Sort, Bubble Sort, Tim Sort	Ranking products, sorting logs by timestamp	Exam merit list, leaderboard sorting
2	Searching Algorithms	Find element in data	Linear Search, Binary Search, Jump Search	Search in arrays, DB indexing logic	Finding a word in dictionary
3	Divide & Conquer	Break problem into smaller sub problems	Merge Sort, Quick Sort, Binary Search	Large-scale computation splitting	Solving a puzzle by parts
4	Recursion Algorithms	Problems requiring repeated sub-steps	DFS, factorial, Fibonacci	Tree traversal, parsing, backtracking	Family tree exploration
5	Backtracking Algorithms	Find all solutions with constraints	N-Queens, Sudoku Solver, Maze solving	Constraint solver, scheduling	Sudoku solving, path finding
6	Greedy Algorithms	Fast local best decisions	Huffman Coding, Prim, Kruskal	Network routing, compression	Selecting cheapest route trip
7	Dynamic Programming (DP)	Optimization with overlapping subproblems	Knapsack, LCS, LIS, Matrix DP	ML feature selection, route planning	Budget planning for best savings
8	Graph Algorithms	Relationship-based problems	BFS, DFS, Dijkstra, Floyd Warshall	Social network, maps, routing	Google Maps route suggestions

9	BFS (Breadth First Search)	Level-order exploration	BFS traversal	Shortest unweighted path	Searching nearest friend in network
10	DFS (Depth First Search)	Deep exploration	DFS traversal	Tree/graph exploration	Maze solving
11	Shortest Path Algorithms	Best path between nodes	Dijkstra, Bellman Ford, A*	Navigation systems, logistics	Delivery route optimization
12	Minimum Spanning Tree (MST)	Cheapest network connection	Kruskal, Prim	Network cable layout, power grid	Connecting cities with minimal roads
13	Topological Sorting	Dependency resolution	Kahn's Algorithm, DFS-based topo sort	Build systems, job scheduling	Course prerequisite order
14	Hashing Algorithms	Fast key-value access	HashMap, HashSet, Rolling Hash	Caching, dictionary storage	Phonebook lookup
15	String Matching Algorithms	Find patterns in text	KMP, Rabin-Karp, Z-Algorithm	Search engine, plagiarism checker	Find word in large document
16	Trie Algorithms	Fast prefix search	Trie, Radix Tree	Auto-suggest, search completion	Mobile keyboard suggestions
17	Compression Algorithms	Reduce data size	Huffman, LZ77, Run-length encoding	File compression, media encoding	ZIP compression
18	Encryption Algorithms	Secure data	AES, RSA, SHA hashing	Secure payments, auth	ATM transaction security
19	Bit Manipulation Algorithms	Efficient low-level operations	XOR tricks, bit masks	OS/embedded systems, fast optimizations	Flags system in permissions
20	Mathematical Algorithms	numeric computations	GCD, Prime sieve, Fast exponentiation	Crypto, scientific computing	Banking interest calculation

21	Number Theory Algorithms	primes, modulo	Sieve of Eratosthenes, Fermat theorem	Encryption, cryptography	Secure password hashing
22	Randomized Algorithms	probabilistic results	Monte Carlo methods, Las Vegas algos	AI simulation, probabilistic models	Casino games / simulation
23	Heuristic Algorithms	approximate best solutions	Hill Climbing, Greedy + heuristics	NP-hard problems	Timetable scheduling
24	Approximation Algorithms	near-optimal solution	Approx knapsack, vertex cover approx	Logistics planning	Truck loading optimization
25	Brute Force Algorithms	try all possibilities	brute search, permutations	cracking/testing, prototypes	Guessing lock combinations
26	Computational Geometry	shape/location problems	Convex Hull, Line intersection	GIS systems, mapping	GPS fence boundary
27	Scheduling Algorithms	task allocation	Round Robin, FCFS, Priority scheduling	OS scheduling, load balancing	CPU scheduling in OS
28	Load Balancing Algorithms	distribute traffic	Round Robin, Least connections	Microservices + API gateways	Splitting traffic across servers
29	Caching Algorithms	improve performance	LRU, LFU, FIFO cache	Redis cache strategy	Keeping recent apps in memory
30	Stream Algorithms	processing live data	Sliding window, Reservoir sampling	fraud detection, IoT analytics	Live stock monitoring
31	Consensus Algorithms	distributed agreement	Paxos, Raft	Distributed DB consistency	Multi ATM sync in banks
32	Data Mining Algorithms	patterns in data	Apriori, FP-Growth	Market basket analysis	Shopping recommendation patterns

33	Rule-Based Algorithms	decision rules	if-else systems, expert systems	early AI systems	Medical symptom decision rule
34	Machine Learning Algorithms	predictive learning	Regression, SVM, Decision tree	predictions, classification	Predict house price
35	Supervised Learning Algorithms	train with labeled data	Linear regression, Random forest	OCR, fraud detection	Spam classification
36	Unsupervised Learning Algorithms	no labels	K-Means, DBSCAN, PCA	customer segmentation	Grouping similar customers
37	Semi-supervised Learning	few labels + many unlabeled	pseudo-labeling	modern ML training	training with limited medical labels
38	Reinforcement Learning Algorithms	agent learns by reward	Q-learning, SARSA, DQN	robotics, games	teaching robot to walk
39	Deep Learning Algorithms	neural learning	CNN, RNN, Transformers	image, NLP	Face recognition
40	CNN Algorithms	vision tasks	ResNet, VGG, MobileNet	object detection, classification	Camera face unlock
41	RNN / LSTM Algorithms	sequence learning	LSTM, GRU	time series forecasting	stock prediction
42	Transformer Algorithms	attention-based learning	BERT, GPT, T5	chatbots, translation	ChatGPT conversations
43	NLP Algorithms	text processing	TF-IDF, Word2Vec, BERT	sentiment analysis, chatbot	customer review analysis
44	Feature Engineering Algorithms	input transformation	scaling, normalization, encoding	ML preprocessing	cleaning raw data
45	Dimensionality Reduction	reduce features	PCA, t-SNE, UMAP	faster ML training	compressing data view

46	Clustering Algorithms	group similar data	KMeans, Hierarchical clustering	segmentation	grouping similar people
47	Anomaly Detection Algorithms	detect outliers	Isolation Forest, Z-score	Fraud detection	credit card fraud
48	Recommendation Algorithms	personalized suggestions	collaborative filtering, matrix factorization	Netflix, Amazon	movie suggestions
49	Computer Vision Algorithms	visual recognition	YOLO, SSD, OpenCV	surveillance	object detection camera
50	Generative Algorithms	generate content	GAN, Diffusion Models, VAE	AI image generation	Deepfake images
51	Optimization Algorithms	best parameter values	Gradient descent, Adam	ML training	minimizing error in model
52	Genetic Algorithms	evolutionary optimization	selection, crossover, mutation	schedule optimization	selecting best breeding pair
53	Swarm Intelligence Algorithms	swarm optimization	PSO, Ant Colony Optimization	shortest route in networks	ants finding shortest path
54	Fuzzy Logic Algorithms	approximate reasoning	fuzzy inference	automation	AC temperature automation
55	Robotics Algorithms	motion/path control	SLAM, A* navigation	robotics navigation	self-driving car navigation
56	Image Processing Algorithms	enhance/filter images	edge detection, thresholding	OCR, scanning apps	photo filter apps
57	Audio Processing Algorithms	speech/music processing	FFT, MFCC	voice assistants	Alexa speech detection
58	Signal Processing Algorithms	analyze signals	Fourier transform, Wavelet	medical signal analysis	ECG analysis

59	Time Series Algorithms	sequential forecasting	ARIMA, LSTM, Prophet	sales prediction	predicting weather
60	Decision Algorithms	choices & branching	Minimax, Alpha-beta pruning	AI game bots	Chess AI moves

✓ Summary (What to remember for Interviews)

- **Java + Python** use heavy **DSA algorithms**: sorting, searching, graphs, DP.
- **AI** uses **learning algorithms**: ML, DL, NLP, optimization.
- **Real-world software systems** use:
 ✓ caching + load balancing + scheduling + graph routing + security algorithms.

If you want, I can also provide:

- ✓ **Algorithm → Data structure needed table**
- ✓ **Algorithm → Time complexity table (Big-O)**
- ✓ **Java + Python code examples for each major algorithm category**

Absolutely ✓

Below is the **MASTER “Algorithm → Time Complexity (Big-O)” Table** covering **Java + Python DSA + AI algorithms**.

✓ Algorithm → Time Complexity Table (Big-O)

1) Sorting Algorithms

Algorithm	Best	Average	Worst	Space	Stable?	Notes / Use
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	✓ Yes	Simple, educational
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$	✗ No	Few swaps
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	✓ Yes	Great for nearly sorted
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	✓ Yes	Reliable + stable
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$	✗ No	Fast in practice

Heap Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$	✗ No	Good worst-case
Counting Sort	$O(n+k)$	$O(n+k)$	$O(n+k)$	$O(k)$	✓ Yes	Only for limited integer range
Radix Sort	$O(nk)$	$O(nk)$	$O(nk)$	$O(n+k)$	✓ Yes	Digit-based sorting
Bucket Sort	$O(n)$	$O(n+k)$	$O(n^2)$	$O(n)$	Depends	Best for uniform distribution
TimSort (Python sort)	$O(n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	✓ Yes	Used in Python, Java (Objects sort)

2) Searching Algorithms

Algorithm	Best	Average	Worst	Space	Notes
Linear Search	$O(1)$	$O(n)$	$O(n)$	$O(1)$	Works on unsorted data
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$	$O(1)$	Requires sorted array
Jump Search	$O(1)$	$O(\sqrt{n})$	$O(\sqrt{n})$	$O(1)$	Requires sorted array
Interpolation Search	$O(1)$	$O(\log \log n)$	$O(n)$	$O(1)$	Works best with uniform distribution
Exponential Search	$O(1)$	$O(\log n)$	$O(\log n)$	$O(1)$	Useful for unknown size/infinite arrays

3) Recursion / Backtracking

Algorithm	Time Complexity	Space	Notes
Factorial recursion	$O(n)$	$O(n)$	recursion stack
Fibonacci (naive recursion)	$O(2^n)$	$O(n)$	inefficient
Fibonacci (DP memoization)	$O(n)$	$O(n)$	efficient

DFS traversal	$O(V+E)$	$O(V)$	recursion/stack
N-Queens	$O(N!)$	$O(N)$	worst-case grows fast
Sudoku solver	$\sim O(9^n)$	$O(n)$	depends on empty cells
Permutations of n elements	$O(n \cdot n!)$	$O(n)$	generate all permutations
Combinations (n choose k)	$O(C(n,k))$	$O(k)$	output sensitive

4) Dynamic Programming (DP)

DP Problem	Time	Space	Notes
0/1 Knapsack (n items, capacity W)	$O(nW)$	$O(nW) / O(W)$	classic DP
Longest Common Subsequence (LCS)	$O(nm)$	$O(nm)$	n,m string lengths
Edit Distance	$O(nm)$	$O(nm)$	used in NLP spellcheck
Matrix Chain Multiplication	$O(n^3)$	$O(n^2)$	optimization DP
Coin Change (min coins)	$O(n \cdot \text{amount})$	$O(\text{amount})$	common interview DP
Longest Increasing Subsequence (DP)	$O(n^2)$	$O(n)$	simple version
LIS (Binary search optimized)	$O(n \log n)$	$O(n)$	most used version

5) Graph Algorithms

Traversal & Core Graph

Algorithm	Time	Space	Notes
BFS	$O(V+E)$	$O(V)$	shortest path unweighted

DFS	$O(V+E)$	$O(V)$	cycle detection, traversal
Topological Sort	$O(V+E)$	$O(V)$	DAG only
Connected Components (DFS/BFS)	$O(V+E)$	$O(V)$	clustering / groups

Shortest Path

Algorithm	Time	Space	Notes
Dijkstra (binary heap)	$O((V+E) \log V)$	$O(V)$	non-negative weights
Dijkstra (array)	$O(V^2)$	$O(V)$	for dense graphs
Bellman-Ford	$O(VE)$	$O(V)$	handles negative weights
Floyd Warshall	$O(V^3)$	$O(V^2)$	all-pairs shortest path
A* Search	$\sim O(E)$ best case	$O(V)$	depends on heuristic

MST

Algorithm	Time	Space	Notes
Kruskal	$O(E \log E)$	$O(V)$	DSU required
Prim (heap)	$O(E \log V)$	$O(V)$	good for sparse graphs
Prim (matrix)	$O(V^2)$	$O(V)$	dense graphs

6) Hashing / Sets / Maps

Data Structure / Algo	Average Time	Worst Time	Notes
HashMap get/put	$O(1)$	$O(n)$	worst due to collisions
HashSet add/contains	$O(1)$	$O(n)$	collision edge case
Rolling Hash (Rabin-Karp)	$O(n+m)$	$O(nm)$	string search

7) String Algorithms

Algorithm	Time	Space	Notes
Naive substring search	$O(nm)$	$O(1)$	slow
KMP	$O(n+m)$	$O(m)$	prefix table
Rabin-Karp	$O(n+m)$ avg	$O(nm)$ worst	hash collisions
Z Algorithm	$O(n+m)$	$O(n+m)$	pattern matching
Trie insert/search	$O(L)$	$O(\text{total chars})$	L = word length

8) Tree Algorithms

Algorithm	Time	Space	Notes
Tree traversal (in/pre/post order)	$O(n)$	$O(h)$	h = height
BST search/insert (balanced)	$O(\log n)$	$O(h)$	AVL/RedBlack
BST search/insert (unbalanced)	$O(n)$	$O(n)$	skew tree
Heap insert/extract	$O(\log n)$	$O(1)$	priority queue
Segment Tree query/update	$O(\log n)$	$O(n)$	range queries
Fenwick Tree update/query	$O(\log n)$	$O(n)$	prefix sums

✓ AI / ML / DL Algorithms → Complexity (Big-O)

In AI, complexity depends on:

n = number of samples, **d** = features, **k** = clusters, **T** = iterations, **p** = parameters

9) Classical Machine Learning

Algorithm	Training Time	Prediction Time	Notes
Linear Regression (normal equation)	$O(nd^2 + d^3)$	$O(d)$	d^3 from matrix inverse
Linear Regression (Gradient Descent)	$O(Tnd)$	$O(d)$	scalable
Logistic Regression (GD)	$O(Tnd)$	$O(d)$	classification
KNN	$O(1)$ (store data)	$O(nd)$	slow prediction
Naive Bayes	$O(nd)$	$O(d)$	very fast
Decision Tree	$O(nd \log n)$	$O(\log n)$	varies
Random Forest	$O(M \cdot nd \log n)$	$O(M \log n)$	M trees
SVM (kernel)	$O(n^2 \text{ to } n^3)$	$O(\#SV \cdot d)$	costly
PCA	$O(\min(nd^2, dn^2))$	$O(d^2)$	matrix decomposition

10) Unsupervised Learning / Clustering

Algorithm	Time	Space	Notes
K-Means	$O(n \cdot k \cdot T \cdot d)$	$O(kd)$	popular clustering
DBSCAN	$O(n \log n)$ avg	$O(n^2)$ worst	density-based
Hierarchical clustering	$O(n^2 \log n)$	$O(n^2)$	expensive

11) Deep Learning / Neural Networks

Model Type	Training Complexity	Inference Complexity	Notes
MLP (Dense NN)	$O(T \cdot n \cdot p)$	$O(p)$	depends on layers
CNN	$O(T \cdot n \cdot F \cdot K^2)$	$O(F \cdot K^2)$	F filters, K kernel
RNN/LSTM	$O(T \cdot n \cdot \text{seq} \cdot h^2)$	$O(\text{seq} \cdot h^2)$	seq length

Transformer	$O(T \cdot n \cdot L^2 \cdot d)$	$O(L^2 \cdot d)$	attention = quadratic
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12) Reinforcement Learning

Algorithm	Time	Space	Notes
Q-Learning (table)	$O(\text{Episodes} \cdot \text{Steps})$	$O(S)$	S
DQN	$O(T \cdot \text{batch} \cdot p)$	$O(p)$	deep Q-learning

★ Interview-Focused Quick List (Most Asked)

Topic	Best Complexity to Remember
Binary Search	$O(\log n)$
Merge Sort	$O(n \log n)$
Quick Sort	$O(n \log n)$ avg, $O(n^2)$ worst
Heap operations	$O(\log n)$
BFS / DFS	$O(V+E)$
Dijkstra	$O((V+E) \log V)$
Floyd Warshall	$O(V^3)$
DP LCS / Edit Distance	$O(nm)$
HashMap avg	$O(1)$

If you want, I can also generate:

- ✓ **One-page printable cheat sheet PDF** (Big-O + tips)
- ✓ **Algorithm → Best DS table** (which DS is used with which algorithm)
- ✓ **Java + Python code templates** for each major category